Marmara Üniversitesi - Mühendislik Fakültesi

DERS İZLEME PROGRAMI

2022-2023 Bahar Yarıyılı

Ders Kodu	Ders Adı	Ders Türü	Haftalık Ders Saati T	Haftalık Ders Saati U	Kredi	ECTS	Kampüs / Haftalık Gün ve Saati / Derslik
MATH259.1	Numerical Methods	Ders	3	0	3,00	4,00	--

Önkoşul Dersi	Önkoşullu Dersi
	ME403 - Computational Methods in Mechanical Engineering / Mühendislik Fakültesi / Makine Mühendisliği (İngilizce) / Lisans ME304 - Mathematical Modeling in Mechanical Engineering / Mühendislik Fakültesi / Makine Mühendisliği (İngilizce) / Lisans ME403 - Computational Methods in Mechanical Engineering / Mühendislik

Dersin Tanımı	Accuracy estimation in numerical methods, error propagation. Nonlinear equations, iteration methods; solution of system of linear equations, ill-conditioned systems; solution by iterative methods. Algebraic eigenvalue problem; interpolation; difference operators; least squares method. Numerical differentiation and integration; differential equations; Initial value problems for ordinary differential equations.
Dersin Kitabı ve/veya Kaynaklar	Numerical Methods for Engineers, S.C. Chapra and R.P. Canale, 7th edition / 2014, McGraw Hill, Applied Numerical Methods with MATLAB for Engineers and Scientists, Steven C. Chapra, 4th edition McGraw Hill, 2018, ,,,,, Elementary Numerical Analysis, 3rd edition, Atkinson, K. E., Han, W., John Wiley & Sons, Inc., 2003, Numerical Analysis, Burden, R.L., and Faires, J. D., 5th edition, PWS Pub. Com., Oxford Univ. Press. 1995, Applied Numerical Analysis, Gerald, Curtis F. and Patrick O. Wheatley, 5th edition, Addison-Wesley, 1994.
Açıklamalar	The homework will be assigned weekly. Each homework is due following week on Thursday by 6pm. Your homework score will be the average of the homework. Each homework will consist of 4-5 questions and should not require more than a few hours. Half of the homework questions will be conceptual. The remaining half will be computational and require performing computations in MATLAB or Excel.

HAFTA	Konular	Kaynak No - İlgili Bölüm
1.Hafta	Numerical Approximations and Floating-point representation	
2.Hafta	Taylor series and Errors, Loss of significance	
3.Hafta	Roots of Equations (Closed methods)@ Bisection, False Position Methods	
4.Hafta	Roots of Equations (Open methods)@ Fixed-point, Newton-Raphson and Secant Methods, System of Non-linear Equations	
5.Hafta	System of Linear Algebraic Equations@ Gauss Elimination	
6.Hafta	LU decomposition, Matrix Inversion	
7.Hafta	Iterative Methods@ Gauss-Seidel, Jacobi Iterations	
8.Hafta	Curve Fitting, Least Squares Regression	
9.Hafta	Nonlinear least squares	
10.Hafta	Polynomial Interpolation	
11.Hafta	Spline Interpolation	
12.Hafta	Numerical Differentiation	
13.Hafta	Numerical Integration, Newton-Cotes Formulae, Romberg's Integration	
14.Hafta	Solution of Ordinary differential equations, Euler method	

	Ölçme Aracı	Adet	Tarih	Başarı Notuna Katkısı (%)	Yarıyıl / Yıliçi Değ. Notuna Katkısı (%)	Sınav Türü
Değerlendirme Araçları	**Yarıyıl / Yılsonu Sınavı**	1	-	40	-	
	Bütünleme Sınavı (varsa)	0	-		-	
	Yarıyıl / Yıliçi Değerlendirme Bilgileri					
	YİDN	1		60	60,0	Ara Sınav
	hw	1		0	0	Ödev



Doç. Dr. Fulya Çallıalp